

Week 4 tutorial. Write the curved-arrow mechanism for every step below.

Reagent 1 is sodium borohydride in methanol at 0 degrees. Reagent 2 is the acid workup, added dropwise after the reaction has finished.

Draw the tetrahedral intermediate and say why the hydride attacks from the less hindered face of the carbonyl.

Question 3. Assign CIP priorities at each stereocentre in the product below and state the configuration you get.

Question 4. Explain why the aldehyde is more electrophilic than the ketone on the previous page, using orbital arguments rather than steric ones.

Question 5. One mark each. Which of these reactions is a redox reaction, which is a substitution, and which is an addition-elimination?

Marking note from the demonstrator. Full marks are for the intermediate and the stereochemical outcome together, not for the arrow pushing on its own.

Further reading. Clayden, chapter 6, and the lecture recording from Tuesday is on the course page under past papers.

Next week: nucleophilic acyl substitution. Read ahead before the seminar because we start from the mechanisms you did here.